Exercise 10.1

Q: 29) $a_n = \frac{1-2n}{1+2n}$

$$a_n = \frac{\cancel{n}\left(\frac{1}{n}-2\right)}{\cancel{n}\left(\frac{1}{n}+2\right)}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\left(\frac{1}{n}-2\right)}{\left(\frac{1}{n}+2\right)}$$

$$= \frac{\frac{1}{\infty}-2}{\frac{1}{\infty}+2} = \frac{0-2}{0+2} = \boxed{-1} \text{ Ans}$$

Q: 33) $a_n = \frac{n^2-2n+1}{n-1}$

$$a_n = \frac{(n-1)^{\cancel{2}}}{\cancel{n-1}} = n-1$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (n-1)$$

$$= \infty - 1$$

$$= \infty \text{ Ans. (Divergent)}$$

Q: 35) $a_n = 1 + (-1)^n$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (1) + \lim_{n\to\infty} (-1)^n$$

$$= 1 + (1, -1)$$

↳ we have two limits

but lim is always unique.

So, the sequence is divergent because it doesnot give a unique value.

Q: 37) $a_n = \left(\frac{n+1}{2n}\right)\left(1 - \frac{1}{n}\right)$

$$a_n = \left(\frac{\cancel{n}}{2\cancel{n}} + \frac{1}{2n}\right)\left(1 - \frac{1}{n}\right)$$

$$a_n = \left(\frac{1}{2} + \frac{1}{2(\infty)}\right)\left(1 - \frac{1}{\infty}\right)$$

$$a_n = \left(\frac{1}{2} + 0\right)\left(1 - 0\right)$$

$$a_n = \frac{1}{2}$$ ~~An~~ (convergent)

42) $a_n = \frac{1}{(0.9)^n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{(0.9)^n}$

$= \frac{1}{\lim_{n\to\infty} (0.9)^n}$

$= \frac{1}{0} = \infty$ (divergent)

45) $a_n = \frac{\sin n}{n}$

$= -1 \leq \sin n \leq 1$

$= -\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$

= The sine function oscillates between -1 and 1.

$= \lim_{n\to\infty} \frac{\sin(n)}{n}$

$= 0$ (convergent)

46) $a_n = \dfrac{\sin^2 n}{n}$

$0 \le \sin^2 n \le 1$

In case of square sine function oscillates from 0 to 1.

so,

$$0 \le \frac{\sin^2 n}{n} \le \frac{1}{n}$$

so, $\lim_{n \to \infty} \dfrac{\sin^2 n}{n} = 0$

47) $a_n = \dfrac{n}{2^n}$ $\left(\dfrac{\infty}{\infty}\right)$ → so, Hopital rule.

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{2^n \ln(2)}$$

$$= \frac{1}{2^{\infty} \ln(2)} = \frac{1}{\infty} = 0$$

$= 0$ (convergent)

49) $a_n = \dfrac{\ln(n+1)}{\sqrt{n}}$ $\left(\dfrac{\infty}{\infty}\right)$

$$= \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{2\sqrt{n}}}$$

$$= \lim_{n\to\infty} \frac{2\sqrt{n}}{n+1}$$

$$= \lim_{n\to\infty} \frac{2\cancel{\sqrt{n}}}{\cancel{\sqrt{n}}\left(\sqrt{n}+\frac{1}{\sqrt{n}}\right)}$$

$$= \lim_{n\to\infty} \frac{2}{\sqrt{n}+\frac{1}{\sqrt{n}}}$$

$$= \frac{2}{\infty+\frac{1}{\infty}} = \frac{2}{\infty} = 0$$ (convergent).

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50) $a_n = \dfrac{\ln n}{\ln 2n}$ $\left(\dfrac{\infty}{\infty}\right)$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{2}{2n}} = \lim_{n\to\infty} \frac{2n}{2n}$$

$$= 1$$

51) $a_n = 8^{1/n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (8)^{1/n}$

$\because \lim_{n\to\infty} (x)^{1/n} = 1$

$= \lim_{n\to\infty} (8)^{1/n}$

$= (8)^{1/\infty} = 8^0 = 1$ (convergent) Ans.

52) $a_n = (0.03)^{1/n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (0.03)^{1/n}$

$= (0.03)^{1/\infty}$

$= (0.03)^0 = 1$ Ans.

53) $a_n = \left(1+\frac{7}{n}\right)^n$

$= \lim_{n\to\infty} \left(1+\frac{7}{n}\right)^n$ $\boxed{\because \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x}$

$= e^7$ Ans.

54) $a_n = \left(1 - \frac{1}{n}\right)^n$

$\because \lim_{n \to \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

so, $= \lim_{n \to \infty} \left(1 + \frac{(-1)}{n}\right)^n$

$= e^{-1}$ (convergent)

55) $a_n = \sqrt[n]{10n}$

$\lim_{n \to \infty} a_n = \lim_{n \to \infty} \left(\sqrt[n]{10n}\right)$

$= \lim_{n \to \infty} (10n)^{1/n}$

$= \lim_{n \to \infty} (10^{1/n} n^{1/n})$

$= \lim_{n \to \infty} (10)^{1/n} \cdot \lim_{n \to \infty} (n)^{1/n}$

$= (10)^{1/\infty} \cdot (\infty)^{1/\infty}$

$= (10)^0 \cdot \infty^0$

$= 1 \cdot 1 = 1$

(Convergent)

56) $a_n = \sqrt[n]{n^2}$

$a_n = \left(\sqrt[n]{n}\right)^2$

$= \lim_{n \to \infty} (n)^{2 \cdot \frac{1}{n}}$

$= \lim_{n \to \infty} (n)^{2/n}$

$= (\infty)^{2/\infty}$

$= \infty^0 = 1$

57) $a_n = \left(\frac{3}{n}\right)^{1/n}$

$$= \frac{\lim_{n \to \infty} (3)^{1/n}}{\lim_{n \to \infty} (n)^{1/n}} = \frac{(3)^0}{(n)^0} = \frac{1}{1}$$

$= 1$

58) $a_n = (n+4)^{1/(n+4)}$

$\because (x)^{1/x} = 1$

So, 1

let $\boxed{x = n+4}$

59) $a_n = \dfrac{\ln n}{n^{1/n}}$

$= \lim_{n\to\infty} \dfrac{\ln n}{n^{1/n}}$

$= \dfrac{\ln(\infty)}{(\infty)^{1/\infty}} = \dfrac{\ln(\infty)}{\infty^0}$

$= \ln(\infty) = \infty$ Ans

(Divergent).

60) $a_n = \ln n - \ln(n+1)$

$= \lim_{n\to\infty} \ln\left(\dfrac{n}{n+1}\right)$

$= \lim_{n\to\infty} \ln\left(\dfrac{\cancel{n}}{\cancel{n}(1+\frac{1}{n})}\right)$

$= \lim_{n\to\infty} \ln\left(\dfrac{1}{1+\frac{1}{n}}\right)$

$= \ln\left(\dfrac{1}{1+\frac{1}{\infty}}\right) = \dfrac{1}{1}$

$= \ln(1) = 0$ Ans

61) $a_n = \sqrt[n]{4^n n}$

$= \lim\limits_{n \to \infty} (4^n n)^{1/n}$

$= \lim\limits_{n \to \infty} [(4^n)^{1/n} \cdot n^{1/n}]$

$= \lim\limits_{n \to \infty} (4 \cdot n^{1/n})$

$= 4 \cdot (\infty)^{1/\infty} = 4 \cdot \infty^0$

$= 4$ Ans. convergent

62) $a_n = \sqrt[n]{3^{2n+1}}$

$= \lim\limits_{n \to \infty} (3^{2n+1})^{1/n}$

$\cancel{= \lim\limits_{n \to \infty} (3^{2n})^{1/n} \cdot (3)^{1/n}}$

$\cancel{= \lim\limits_{n \to \infty} (3)^{2n/n} \cdot (3)^{1/n}}$

$\cancel{= (3)^{\infty/\infty} \cdot (3)^{1/\infty}}$

$\cancel{= 1 \cdot 1}$

$= \lim\limits_{n \to \infty} \sqrt[n]{(3)^{n(2+1/n)}}$

$= \lim\limits_{n \to \infty} 3^{n(2+1/n) \times \frac{1}{n}}$

$= \lim\limits_{n \to \infty} (3^2 \cdot 3^{1/n})$

$= 3^2 \cdot (3)^{1/\infty}$

$= 9 \cdot 1 = 9$ Ans.

63) $a_n = \dfrac{n!}{n^n}$

$\lim\limits_{n \to \infty} a_n = \lim\limits_{n \to \infty} \dfrac{n!}{n^n}$

$\because \quad 0 < \dfrac{n!}{n^n} < \dfrac{1}{n}$

Limit is zero so zero is finite number.

$= \lim_{n\to\infty} \dfrac{n!}{n^n} = 0$ (converges)

64) $\dfrac{(-4)^n}{n!}$

Theorem: $\because \lim_{n\to\infty} \dfrac{x^n}{n!} = 0$

so, $(x=-4)$ ... (converges).

65) $a_n = \dfrac{n!}{10^{6n}}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\dfrac{1}{\dfrac{10^{6n}}{n!}} \right)$

$= \lim_{n\to\infty} \dfrac{1}{\left(\dfrac{(10^6)^n}{n!}\right)} \quad \because \dfrac{x^n}{n!} = 0$

$\frac{1}{0} = \infty$ (diverges) so.

66) $a_n = \frac{n!}{2^n 3^n}$

$= \lim_{n \to \infty} \frac{n!}{2^n 3^n} \Rightarrow \lim_{n \to \infty} \frac{n!}{6^n}$

$= \lim_{n \to \infty} \left(\frac{1}{\frac{(6)^n}{n!}} \right)$

$= \frac{1}{\lim_{n \to \infty} \frac{(6)^n}{n!}} = \frac{1}{0} = \infty$ D

67) $a_n = \left(\frac{1}{n}\right)^{\frac{1}{\ln(n)}}$

$= \lim_{n \to \infty} \left(\frac{1}{n}\right)^{\frac{1}{\ln(n)}}$

let $y = \left(\frac{1}{n}\right)^{\frac{1}{\ln(n)}}$

Taking ln on b/sides.

$$\ln y = \ln\left(\frac{1}{n}\right)^{\frac{1}{\ln(n)}}$$

$$\ln y = \frac{1}{\ln(n)}\left(\ln\left(\frac{1}{n}\right)\right)$$

$$y = e^{\left(\frac{1}{\ln(n)}(\ln(1)-\ln(n))\right)}$$

$$y = e^{\left(\frac{1}{\ln(n)}(0-\ln(n))\right)}$$

$$y = e^{-\frac{\ln(n)}{\ln(n)}}$$

$$y = e^{-1}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}(y)$$

$$= \lim_{n\to\infty}(e^{-1})$$

$$= e^{-1} \text{ (converges)}$$

68) $a_n = \ln\left(1+\frac{1}{n}\right)^n$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \ln\left(1+\frac{1}{n}\right)^n$$

$$= \ln\left(\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\right)$$

$$= \ln(e) = 1$$

(converges)

69) $a_n = \left(\frac{3n+1}{3n-1}\right)^n$

$= \lim_{n\to\infty} \left(\frac{3n+1}{3n-1}\right)^n$

let $y = \left(\frac{3n+1}{3n-1}\right)^n$

$\ln y = \ln\left(\frac{3n+1}{3n-1}\right)^n$

$\ln y = n \ln\left(\frac{3n+1}{3n-1}\right)$

$\ln y = n(\ln(3n+1) - \ln(3n-1))$

$y = (e)^{\frac{\ln(3n+1)-\ln(3n-1)}{1/n}}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (e)^{\frac{\ln(3n+1)-\ln(3n-1)}{1/n}}$

Taking derivative.

$= \lim_{n\to\infty} \left(e\right)^{\frac{\frac{3}{3n+1}-\frac{3}{3n-1}}{-\frac{1}{n^2}}}$

$= \lim_{n\to\infty} (e)^{\frac{\frac{3(\cancel{3n}-1-\cancel{3n}-1)}{(3n)^2-(1)^2}}{-1/n^2}}$

$$= \lim_{n\to\infty} (e)^{\frac{6n^2}{9n^2-1}}$$

$$= \lim_{n\to\infty} (e)^{\frac{6n^2}{n^2(9-1/n^2)}}$$

$$= \lim_{n\to\infty} (e)^{\frac{6}{9-1/n^2}}$$

$$= (e)^{\frac{6}{9-1/\infty}}$$

$$= e^{6/9}$$

$$= e^{2/3} \text{ (converges)}$$

70) $$a_n = \left(\frac{n}{n+1}\right)^n$$

~~let y~~ $$= \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^n$$

let $$y = \left(\frac{n}{n+1}\right)^n$$

$$\ln y = n(\ln(n) - \ln(n+1))$$

$$\ln y = \frac{\ln(n) - \ln(n+1)}{1/n}$$

$$y = (e)^{\frac{\ln(n)-\ln(n+1)}{1/n}}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (e)^{\frac{\ln(n)-\ln(n+1)}{1/n}}$$

$$= \lim_{n\to\infty} (e)^{\frac{\frac{1}{n}-\frac{1}{n+1}}{-\frac{1}{n^2}}}$$

$$= \lim_{n\to\infty} (e)^{\frac{\frac{\cancel{n}+1-\cancel{n}}{n^2+n}}{-1/n^2}}$$

$$= \lim_{n\to\infty} (e)^{\frac{-\cancel{n^2}}{\cancel{n^2}(1+\frac{1}{n})}}$$

$$= \lim_{n\to\infty} (e)^{-\frac{1}{1+1/n}}$$

$$= (e)^{-\frac{1}{1+\frac{1}{\infty}}} = e^{-1} \text{ (converges)}$$

71) (imp) $a_n = \left(\frac{x^n}{2n+1}\right)^{1/n}, \quad x>0$

$$= \lim_{n\to\infty} \left(\frac{x}{(2n+1)^{1/n}}\right)$$

$$= x \cdot \lim_{n\to\infty} \left(\frac{1}{(2n+1)^{\cancel{1/n}}}\right)^{1/n} \quad \text{--- (i)}$$

~~$\ln a_n = x$~~ let $y = \left(\frac{1}{2n+1}\right)^{1/n}$

$$\ln y = \frac{1}{n} \ln\left(\frac{1}{2n+1}\right)$$

$$\ln y = \frac{\ln(1) - \ln(2n+1)}{n}$$

$$y = (e)^{-\frac{\ln(2n+1)}{n}}$$

$$\lim_{n\to\infty} a_n = x \cdot \lim_{n\to\infty} (e)^{-\frac{\ln(2n+1)}{n}}$$

$$= x \cdot \lim_{n\to\infty} (e)^{-\frac{2}{2n+1}}$$

$$= x(e)^{-\frac{2}{2(\infty)+1}}$$

$$= x(e)^0 = x \cdot 1$$

$$= x \text{ (converges) } \quad x > 0$$

72) $a_n = \left(1 - \frac{1}{n^2}\right)^n$

$$\lim_{n\to\infty} \ln a_n = \lim_{n\to\infty} \ln\left(1 - \frac{1}{n^2}\right)^n$$

$$= \lim_{n\to\infty} n \ln\left(1 - \frac{1}{n^2}\right)$$

$$\ln a_n = \lim_{n\to\infty} \frac{\ln\left(1 - \frac{1}{n^2}\right)}{1/n}$$

$$a_n = \lim_{n\to\infty} (e)^{\left(\frac{\ln(1 - 1/n^2)}{1/n}\right)}$$

$$a_n = \lim_{n\to\infty} (e)^{\frac{\frac{1}{1-1/n^2} \times \frac{2}{n^3}}{-1/n^2}}$$

$$a_n = \lim_{n\to\infty} (e)^{-\frac{\frac{2}{n^3} \times n^2}{1-1/n^2}} \Rightarrow \lim_{n\to\infty} (e)^{\frac{-2/\cancel{n}}{\frac{n^2-1}{n^2}}}$$

$$a_n = \lim_{n\to\infty} (e)^{-\frac{2n}{n^2-1}}$$

$$a_n = \lim_{n\to\infty} (e)^{-\frac{2\cancel{n}}{\cancel{n}(n-1/n)}}$$

$$a_n = (e)^{-\frac{2}{\infty}} = (e)^0 = 1 \text{ (converges)}$$

73) $a_n = \dfrac{3^n \cdot 6^n}{2^{-n} \cdot n!}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3^n \cdot 6^n \cdot 2^n}{n!}$$

$$a_n = \lim_{n\to\infty} \left(\frac{36^n}{n!}\right) \quad \boxed{\because \lim_{n\to\infty} \frac{(x)^n}{n!} = 0}$$

$$a_n = 0 \text{ (converges)}$$

74) $a_n = \dfrac{(10/11)^n}{(9/10)^n + (11/12)^n}$

Multiply and divide by $\left(\frac{12}{11}\right)^n$ ~~on b/s~~

$$a_n = \frac{(10/11)^n (12/11)^n}{(12/11)^n (9/10)^n + (\cancel{11}/\cancel{12})^n (\cancel{12}/\cancel{11})^n}$$

$$a_n = \frac{(120/121)^n}{(108/110)^n + 1} = 0 \text{ (converges)}$$